

RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

RA8P1 IO allocation



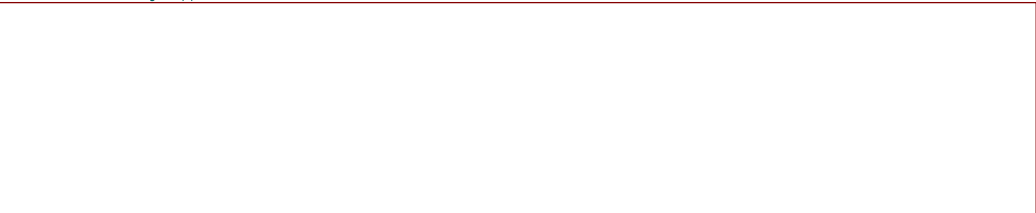
File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator



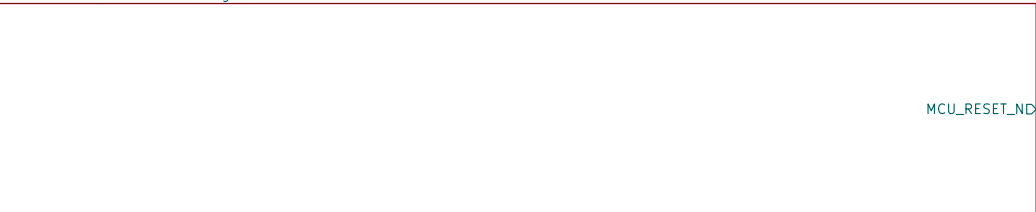
File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

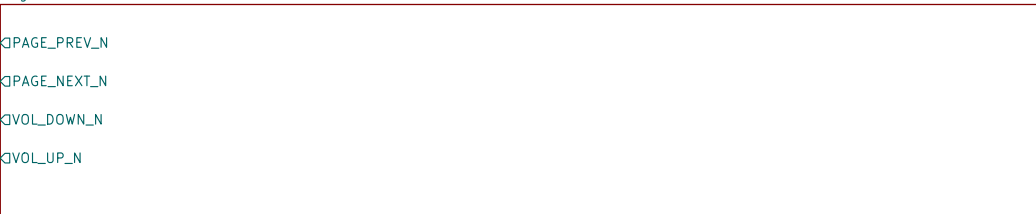
RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

Page and volume buttons



File: user_controls.kicad_sch

ESP32-C6 radio



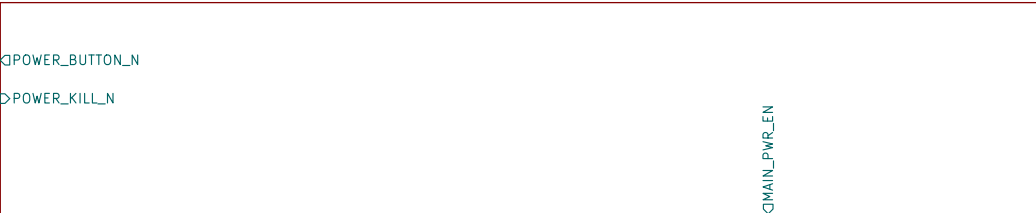
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64 MiB SDRAM



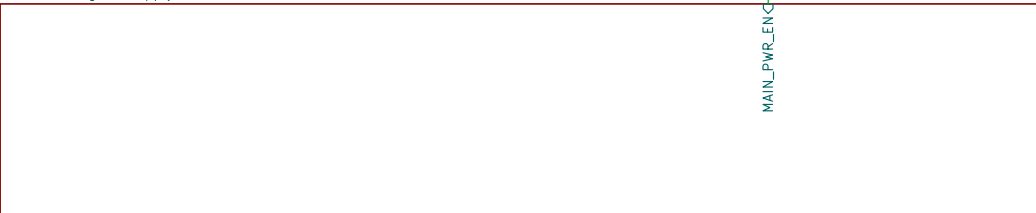
File: sdram.kicad_sch

Power button and source control



File: power_button.kicad_sch

Main 3V3 digital supply



File: main_3v3_supply.kicad_sch

Sheet: /
File: ereader_rev1.kicad_sch

Title: RA8P1 e-reader – schematic index

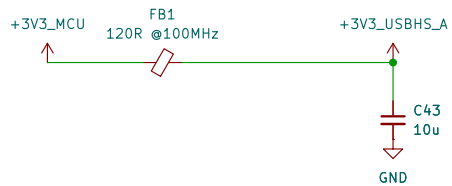
Size: A3 Date: 2026-09-05

KiCad E.D.A. 10.0.5

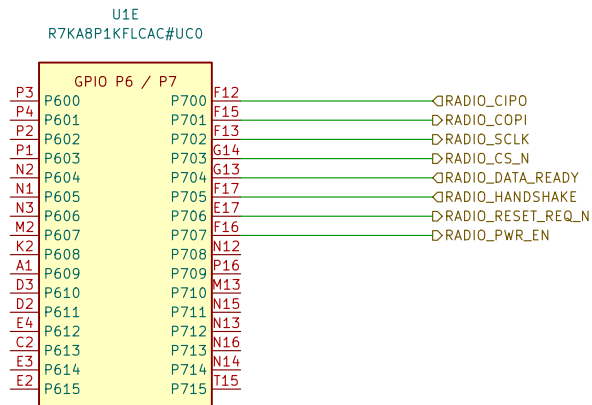
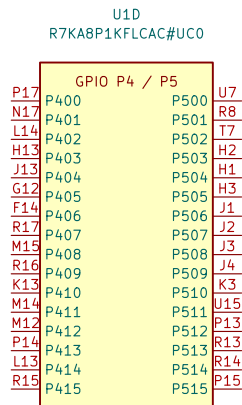
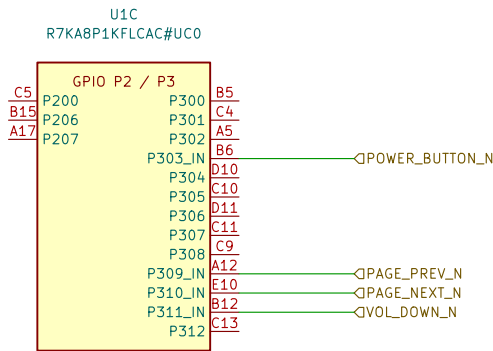
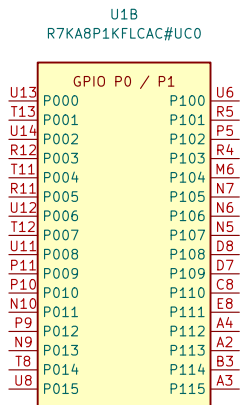
Rev: 0.3 DRAFT

Id: 1/10

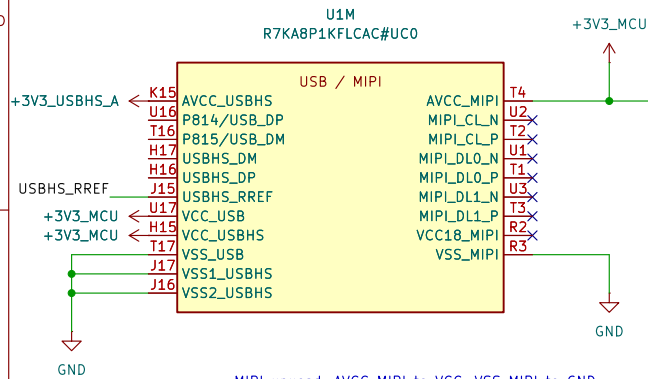
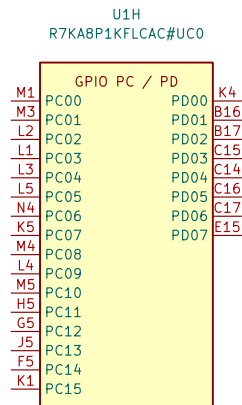
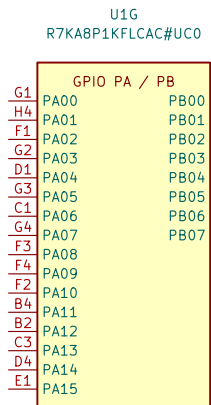
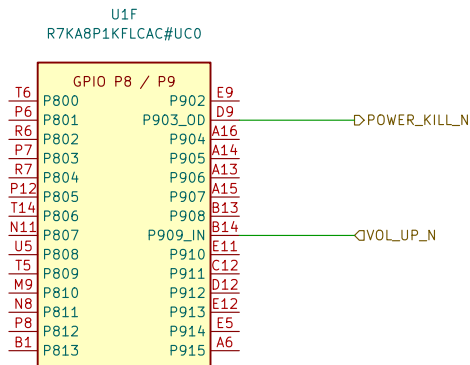
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



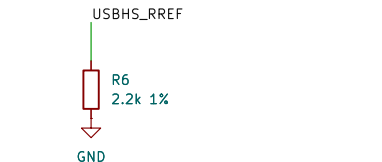
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



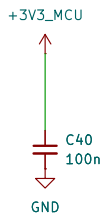
RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



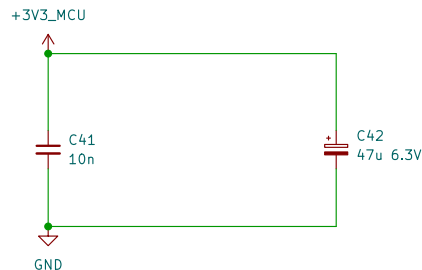
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90.110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9.11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6.56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90.110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

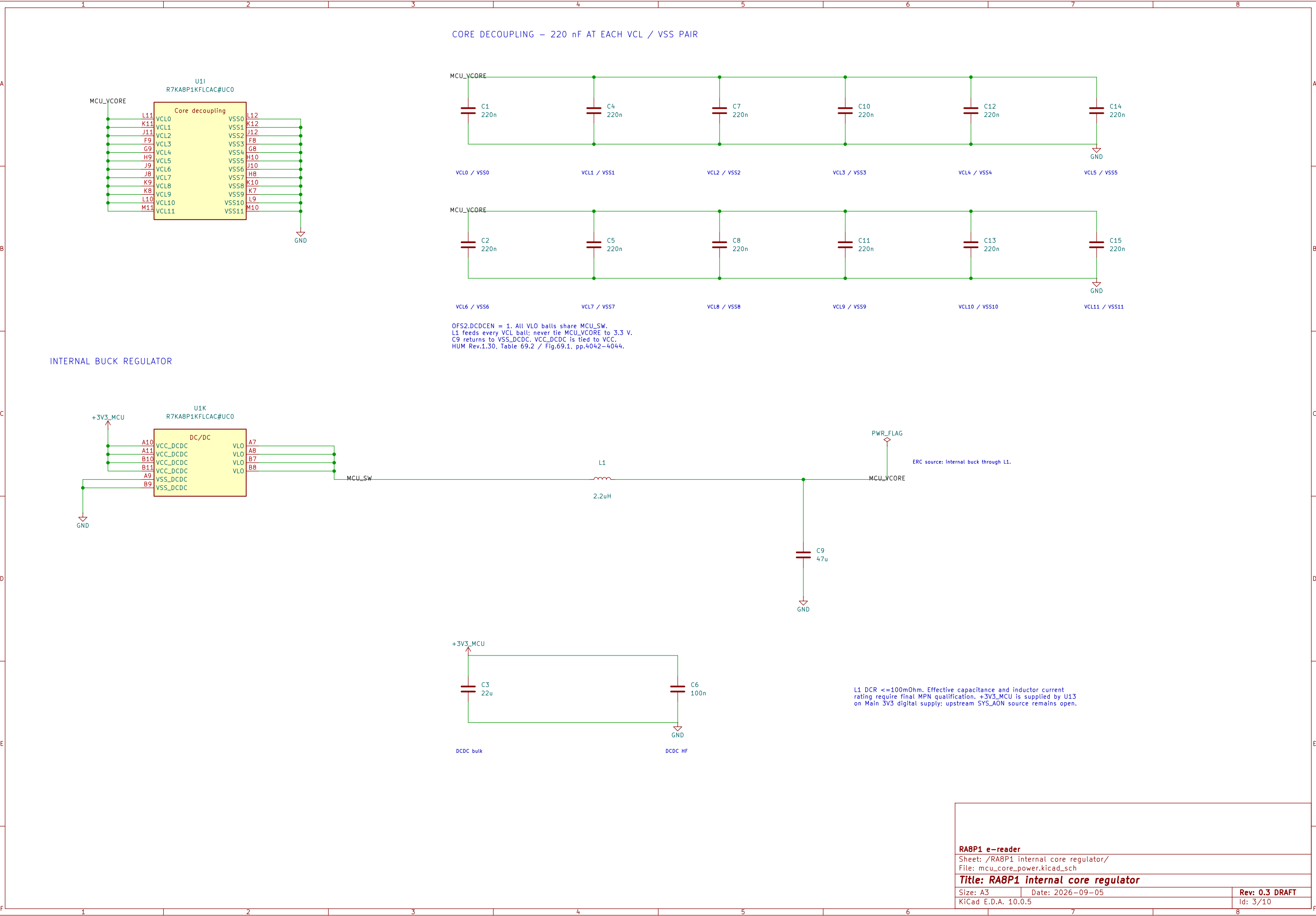
RA8P1 e-reader

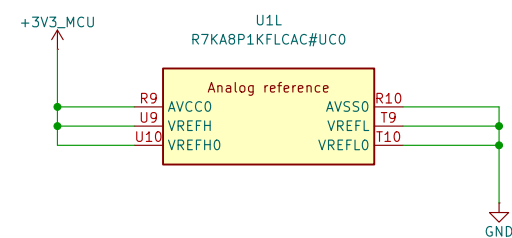
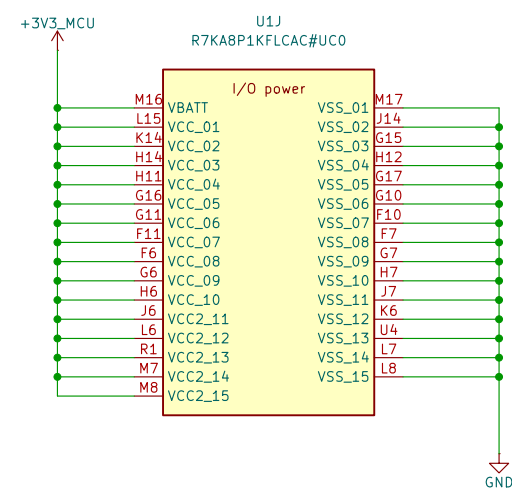
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File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

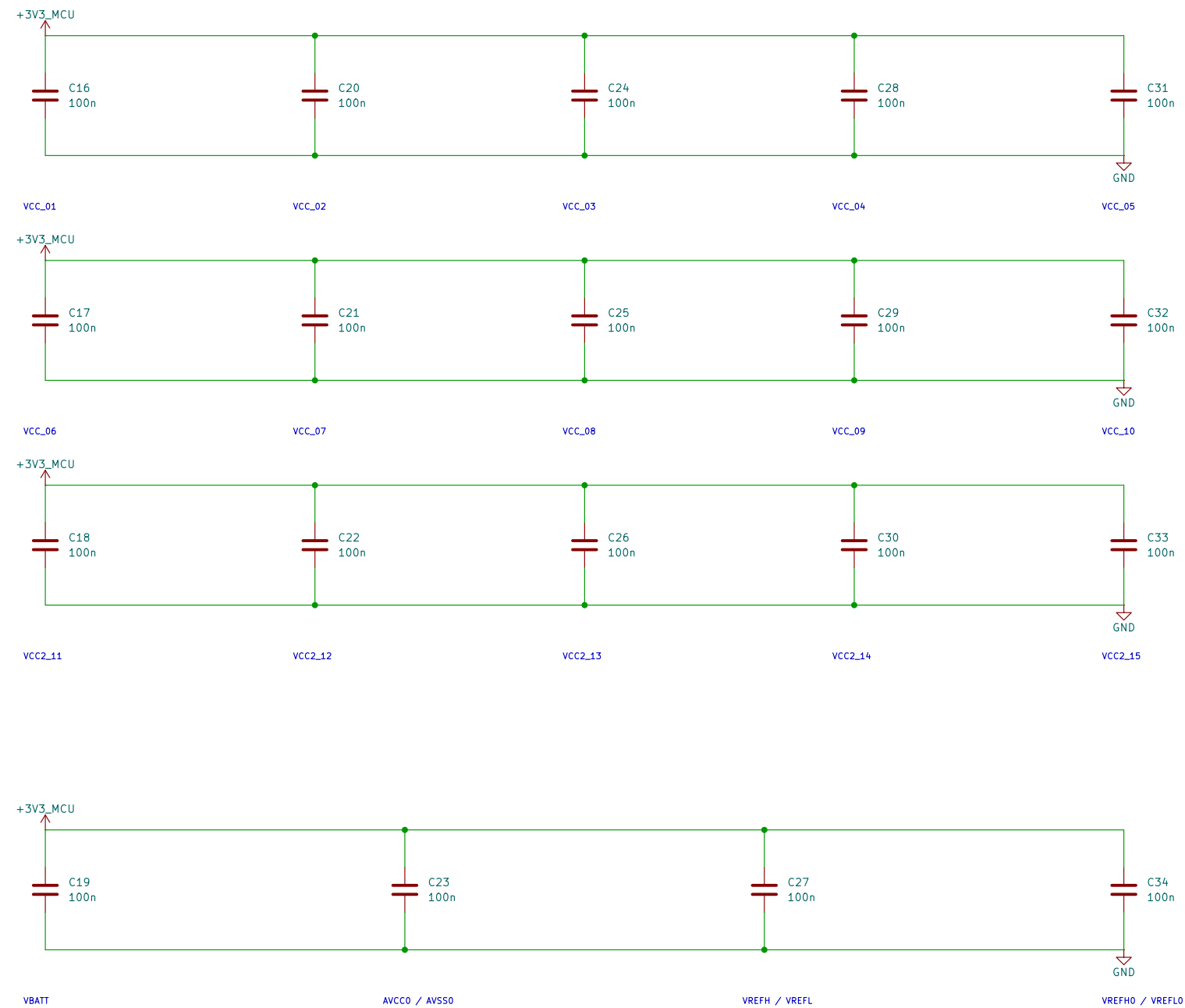
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Rev: 0.3 DRAFT
Id: 2/10

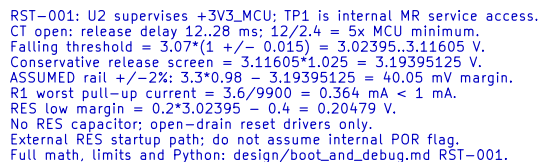
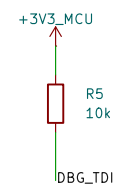
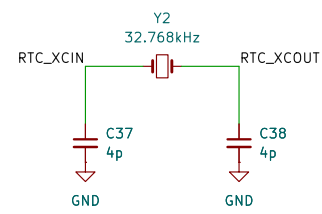
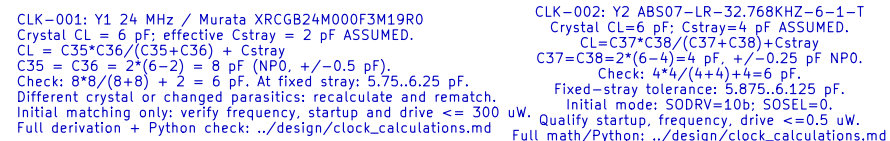




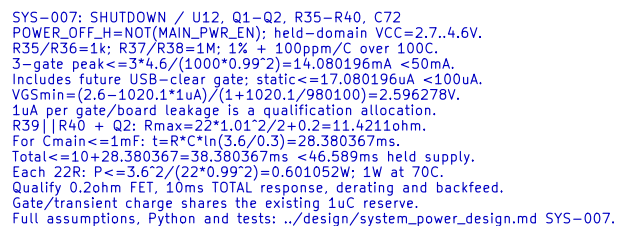
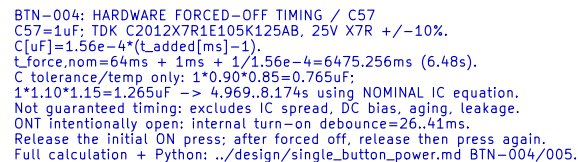
I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



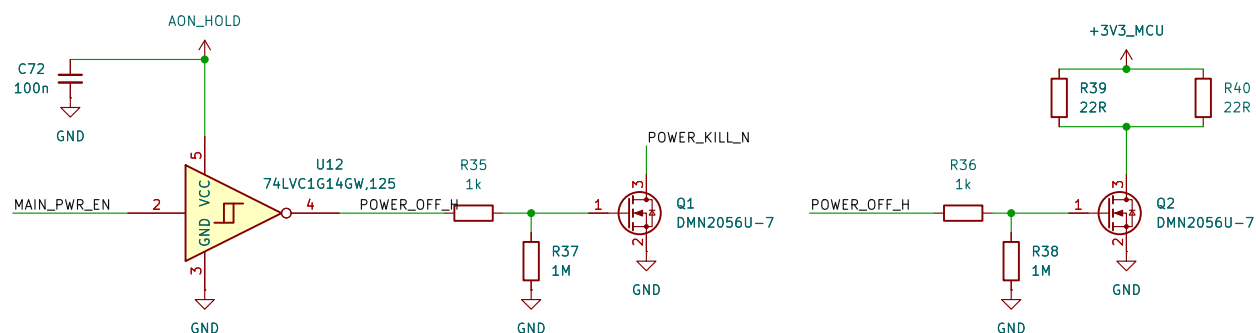
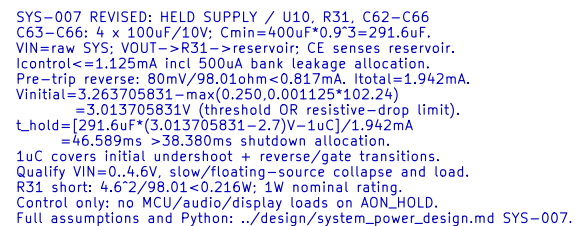
VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIP1 supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

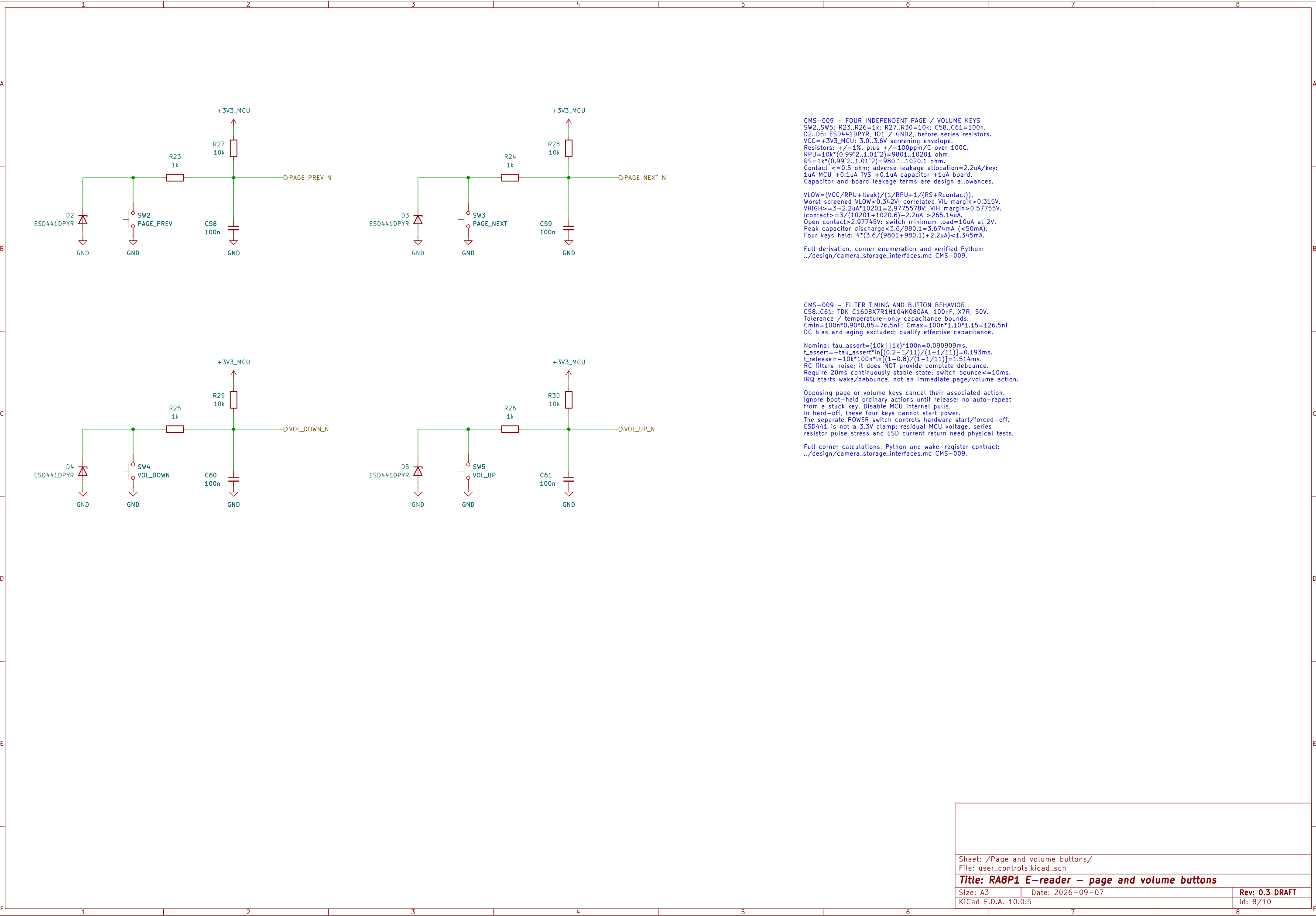


References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3–4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

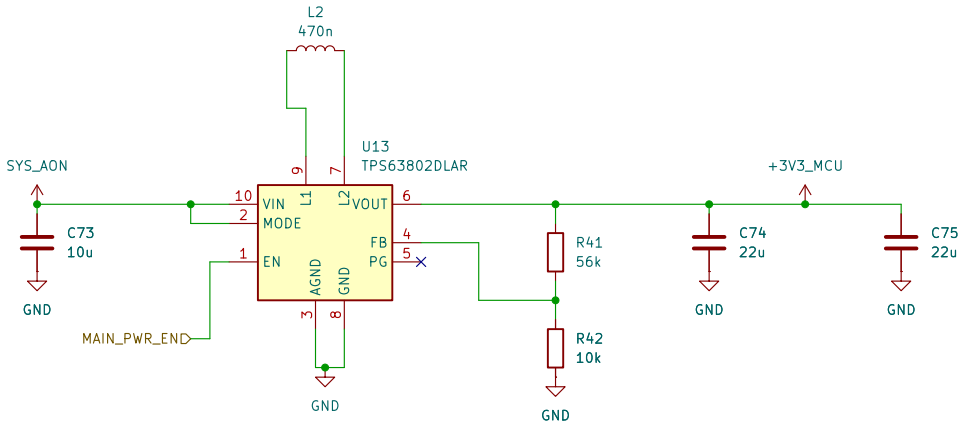


ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





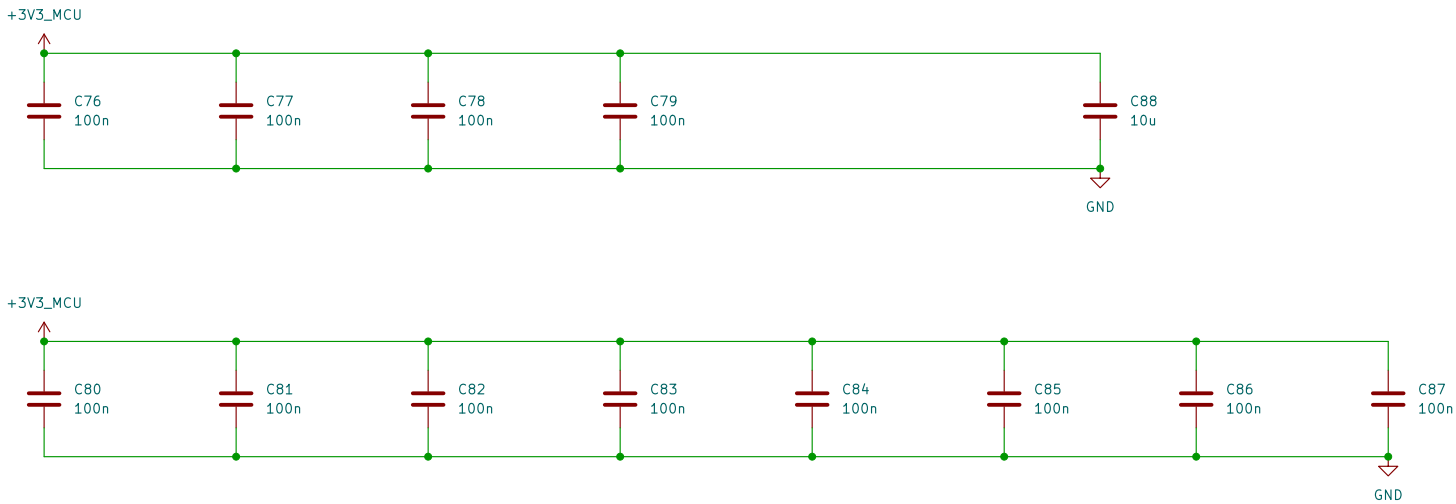
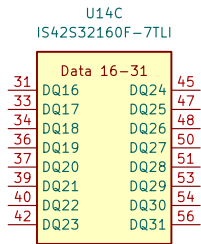
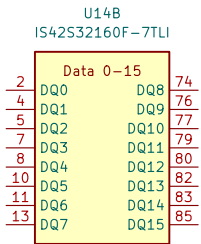
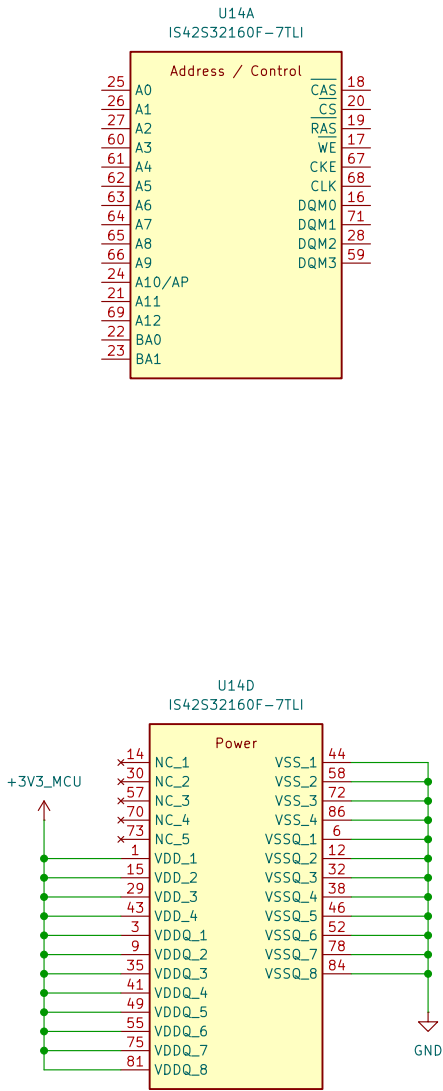
MAIN DIGITAL 3.3 V SUPPLY
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS
U13 TPS63802DLAR; MODE=VIN, fixed PWM.
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.
Vnom=0.500*(1+56k/10k)=3.300V.
Idivider=3.300/(56k+10k)=50uA.
For 100C resistor–temperature change:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vmin=0.495*(1+56k*fLo/(10k*fHi))–100nA*56k*fHi
=3.242044V.
Vmax=0.505*(1+56k*fHi/(10k*fLo))+100nA*56k*fHi
=3.358485V.
Static screen only; ripple, load steps and drift still require validation.
C73: CIN_model=10u*0.90*0.85*0.60=4.590uF >=4uF.
C74+C75: COUT_model=2*22u*0.90*0.85*0.60=20.196uF >=7uF.
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS
Main continuous allocation: 0.750 MCU + 0.500 radio
+ 0.250 SDRAM + 0.100 NOR + 0.050 logic = 1.650A.
These are qualification allocations, not measured maximum currents.
SD, camera, audio, display/touch/frontlight need separate SYS supplies.
Pout=3.358485094*1.65=5.541500W.
VIN>=3.20V allocation; raw trip min=3.263705831V.
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.
At ASSUMED eta=85%: Iin=Pout/(eta*3.20)=2.037316A.
Ploss=Pout*(1/eta–1)=0.977912W.
Reference rise=Ploss*81C/W=79.210859C; NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.400 other = 1.730A.
Radio on would give 2.230A > nominal 2A; keep radio OFF through start/wake.
Largest wake–reference subtotal=1.270+0.400=1.670A.
DCDC figures are reference–only; non–DCDC start and extra charging excluded.
Additional charging: I=Cadditional*dV/dt; qualify the actual ramp.
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.
No externally driven +3V3_MCU: forced PWM can sink reverse current.
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: ../design/power_decoupling.md PWR–003;
source–loss shutdown: ../design/system_power_design.md SYS–007.

64 MiB SDRAM | ISSI IS42S32160F-7TLI
POWER/BYPASS PHASE: signal interface is not yet connected.
Address, data, command pulls and source clock link: see CMS-010.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.